

AWS Secure Cloud Storage

Security, Monitoring, Detection & Alerting Lab

Final Project Documentation

Created By - Tanmay Patil

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1. Project Overview

The project is a controlled AWS security lab built around a private Amazon S3 bucket. The goal was to protect stored data, apply least-privilege access, collect security activity, detect selected high-risk actions, send alerts, and perform additional event analysis with Python.

The completed implementation uses AWS IAM for access control, Amazon S3 for storage, AWS CloudTrail for activity logging, Amazon CloudWatch Logs for centralized log storage, CloudWatch Metric Filters and Alarms for detection, Amazon SNS for email notifications, and a local Python analyzer using boto3 and pandas for additional analysis.

Area	Implementation
Storage security	Private S3 bucket, Block Public Access, HTTPS-only access, SSE-S3 encryption, versioning
Access control	Developers and Auditors groups with different permissions; unauthorized user denied
Audit logging	CloudTrail management events and S3 object-level data events during testing
Log processing	CloudTrail events delivered to /cloudtrail/cloudsec-lab
Detection	DeleteBucket, PutBucketAcl, and ConsoleLogin without MFA
Alerting	CloudWatch alarms connected to SNS email notifications
Additional analysis	Python analyzer for sensitive actions and high API-call volume

1.1 What was actually completed

- Configured the AWS lab environment and verified the administrative identity.
- Secured an S3 bucket against public access and non-HTTPS access.
- Enabled S3 versioning and server-side encryption.
- Created IAM groups and validated allowed and denied access.
- Enabled S3 object-level CloudTrail data events.
- Forwarded CloudTrail logs into CloudWatch Logs.
- Created three CloudWatch detection filters and alarms.
- Validated all three alarm paths and received SNS email notifications.
- Ran a Python analyzer that produced HIGH and MEDIUM findings.
- Decommissioned the lab resources after evidence collection.

2. Objectives and Scope

2.1 Objectives

- Keep the S3 data bucket private and prevent public exposure.
- Require encrypted storage and HTTPS-based access.
- Apply least-privilege IAM permissions.
- Record relevant AWS activity with CloudTrail.
- Detect selected security-relevant events.
- Generate automated email alerts.
- Analyze recent CloudTrail activity with a small Python script.
- Validate the controls with deliberate security tests.

2.2 Scope

The lab was intentionally kept small. It focused on S3, IAM, CloudTrail, CloudWatch, SNS, AWS CLI usage, and a local Python analyzer. No additional production services added.

3. Architecture

The architecture follows a simple security-monitoring chain: identities are controlled by IAM, authorized requests reach S3, CloudTrail records activity, CloudWatch Logs stores the events, metric filters identify selected patterns, alarms evaluate the metrics, and SNS sends notifications. The Python analyzer reads CloudWatch Logs separately for additional analysis.

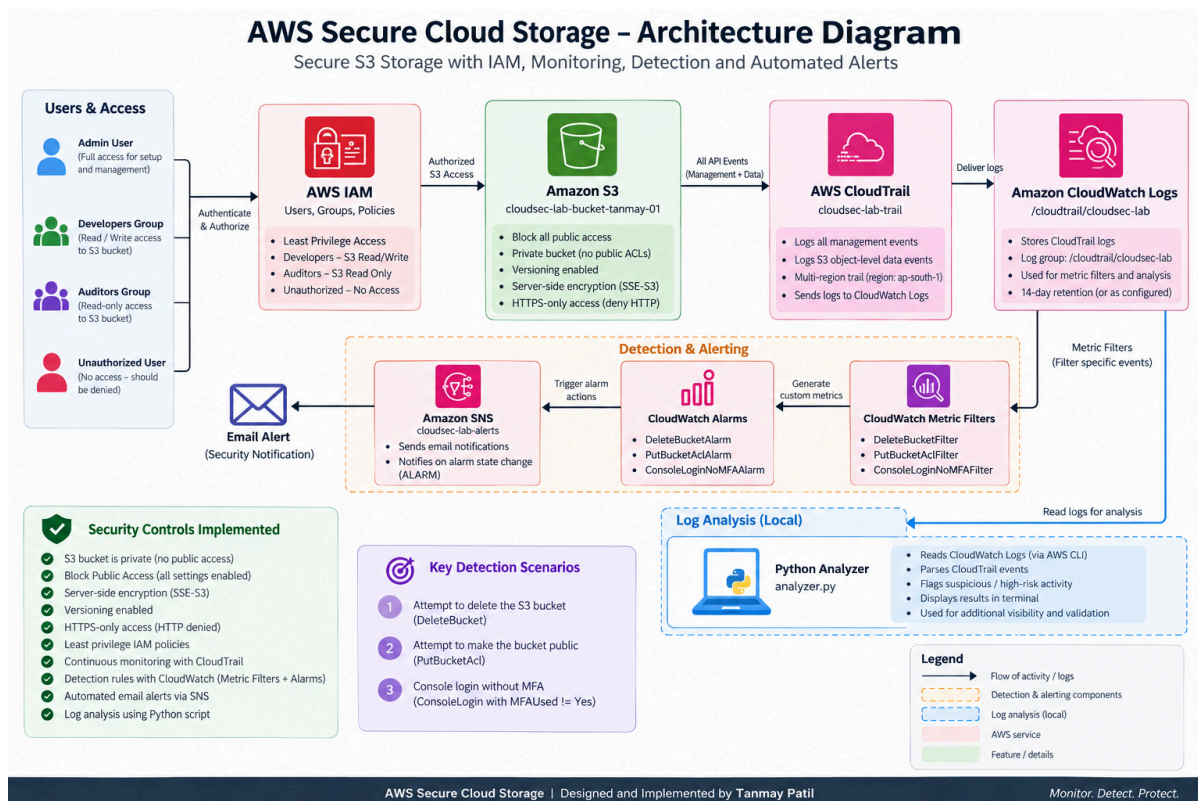
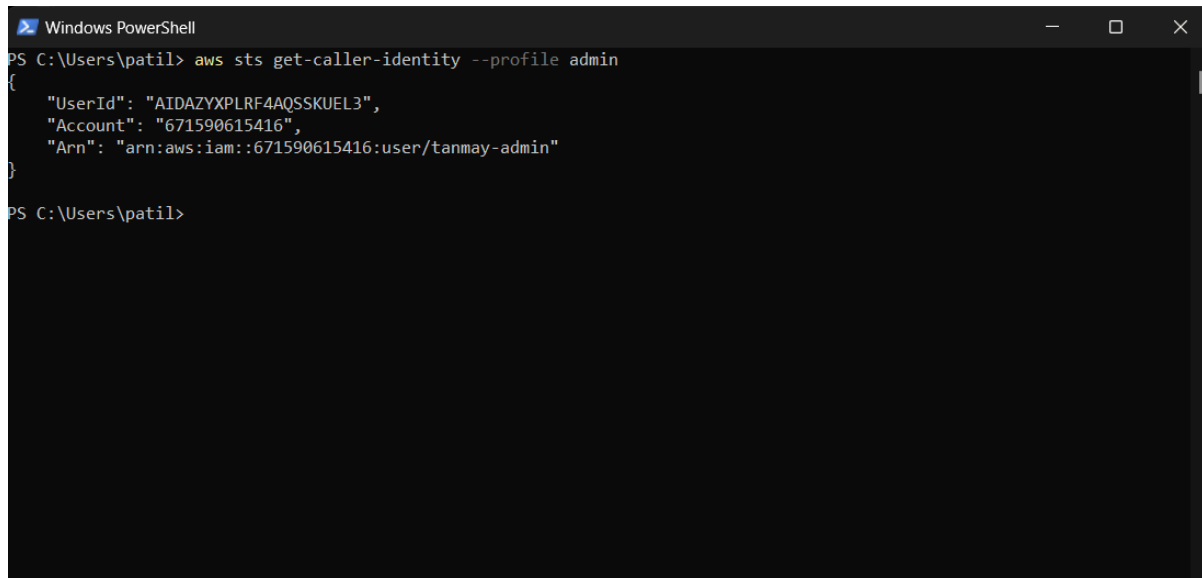


Figure 1. Architecture Diagram

AWS Secure Cloud Storage

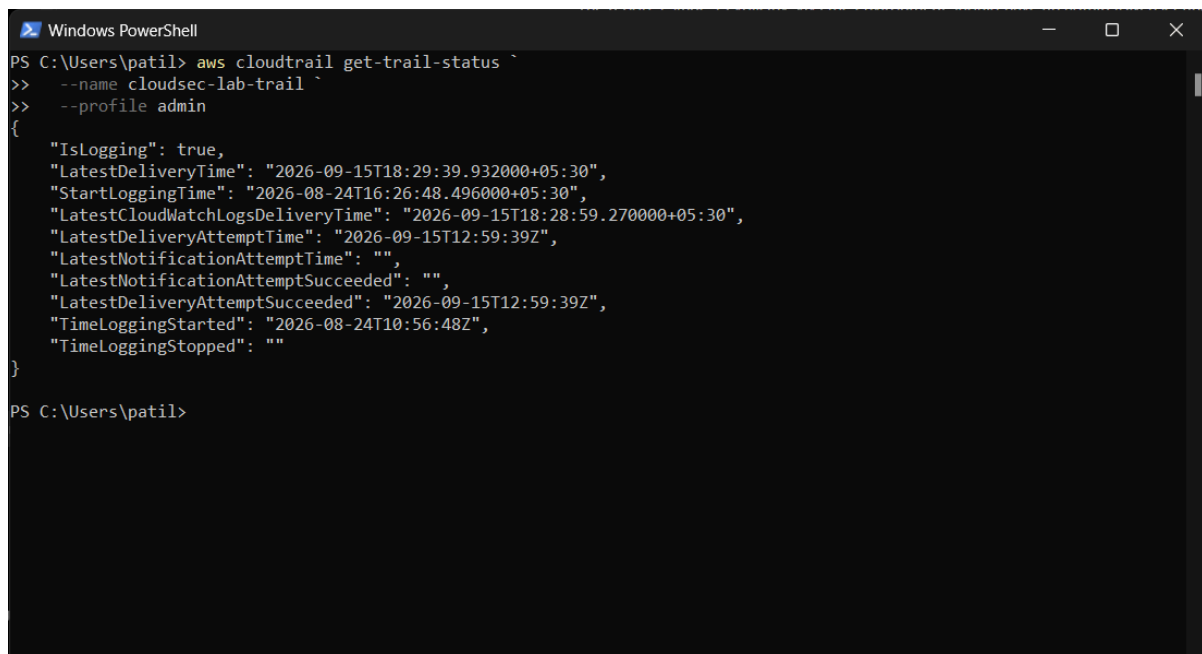
4. Environment Setup

The lab used the Mumbai region (ap-south-1). A dedicated IAM administrative user was used for the lab instead of performing the work as the root identity. CloudTrail was created and verified as actively logging.



```
Windows PowerShell
PS C:\Users\patil> aws sts get-caller-identity --profile admin
{
  "UserId": "AIDAZYXPLRF4AQSSKUEL3",
  "Account": "671590615416",
  "Arn": "arn:aws:iam::671590615416:user/tanmay-admin"
}
PS C:\Users\patil>
```

Figure 2. AWS caller identity verification for the lab administrative user.



```
Windows PowerShell
PS C:\Users\patil> aws cloudtrail get-trail-status `
>> --name cloudsec-lab-trail `
>> --profile admin
{
  "IsLogging": true,
  "LatestDeliveryTime": "2026-09-15T18:29:39.932000+05:30",
  "StartLoggingTime": "2026-08-24T16:26:48.496000+05:30",
  "LatestCloudWatchLogsDeliveryTime": "2026-09-15T18:28:59.270000+05:30",
  "LatestDeliveryAttemptTime": "2026-09-15T12:59:39Z",
  "LatestNotificationAttemptTime": "",
  "LatestNotificationAttemptSucceeded": "",
  "LatestDeliveryAttemptSucceeded": "2026-09-15T12:59:39Z",
  "TimeLoggingStarted": "2026-08-24T10:56:48Z",
  "TimeLoggingStopped": ""
}
PS C:\Users\patil>
```

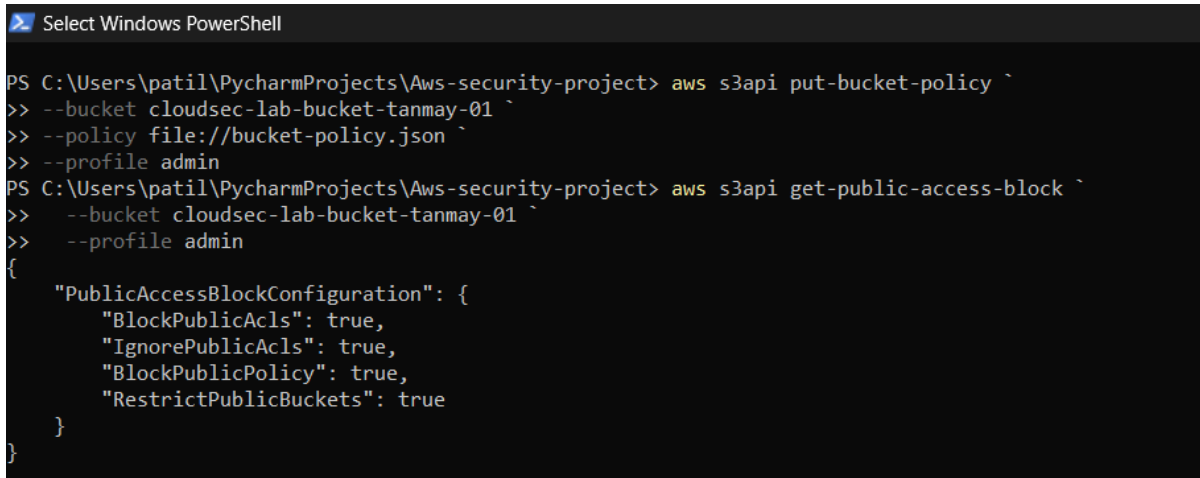
Figure 3. CloudTrail trail status showing active logging and successful delivery activity.

5. S3 Security Controls

The main S3 bucket was configured as a private storage location. The security controls were validated rather than only enabled in the console or through the CLI.

5.1 Block Public Access

All four S3 Block Public Access settings were enabled.



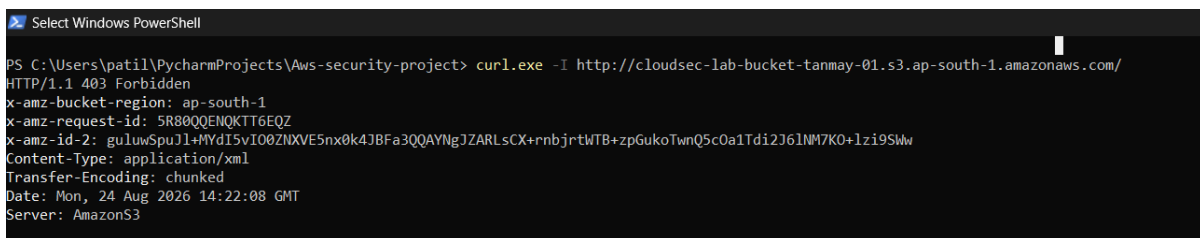
```
Select Windows PowerShell

PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3api put-bucket-policy `
>> --bucket cloudsec-lab-bucket-tanmay-01 `
>> --policy file://bucket-policy.json `
>> --profile admin
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3api get-public-access-block `
>> --bucket cloudsec-lab-bucket-tanmay-01 `
>> --profile admin
{
  "PublicAccessBlockConfiguration": {
    "BlockPublicAcls": true,
    "IgnorePublicAcls": true,
    "BlockPublicPolicy": true,
    "RestrictPublicBuckets": true
  }
}
```

Figure 4. S3 Block Public Access configuration showing all four settings enabled.

5.2 HTTPS-only access

An HTTP request was deliberately sent to the S3 endpoint. The request returned HTTP 403 Forbidden, confirming that the HTTPS-only policy prevented non-TLS access.



```
Select Windows PowerShell

PS C:\Users\patil\PycharmProjects\Aws-security-project> curl.exe -I http://cloudsec-lab-bucket-tanmay-01.s3.ap-south-1.amazonaws.com/
HTTP/1.1 403 Forbidden
x-amz-bucket-region: ap-south-1
x-amz-request-id: 5R80QQENQKTT6EQZ
x-amz-id-2: guluwSpUll+MYdISvI00ZNXVE5nx0k4JBFA3QQAYNgJJZARLsCX+rnbjrtWTB+zpGukoTwnQ5c0a1TdI2J6lNM7K0+lzi9Sww
Content-Type: application/xml
Transfer-Encoding: chunked
Date: Mon, 24 Aug 2026 14:22:08 GMT
Server: AmazonS3
```

Figure 5. HTTP access test returning 403 Forbidden.

5.3 Encryption and versioning

A test object was uploaded and checked with S3 metadata. The object reported AES256 server-side encryption, and the version listing showed an actual version ID.

```
Select Windows PowerShell
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 cp test.txt s3://cloudsec-lab-bucket-tanmay-01/test.txt `
>> --profile admin
upload: .\test.txt to s3://cloudsec-lab-bucket-tanmay-01/test.txt
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3api head-object `
>> --bucket cloudsec-lab-bucket-tanmay-01 `
>> --key test.txt `
>> --profile admin
{
  "AcceptRanges": "bytes",
  "LastModified": "2026-08-24T14:24:13+00:00",
  "ContentLength": 31,
  "ETag": "\"2da08c41858a1281472469b4869c1a1c\"",
  "VersionId": "P4_6gDlRaeVhy7gelFem2SmTapBHzQ0v",
  "ContentType": "text/plain",
  "ServerSideEncryption": "AES256",
  "Metadata": {}
}
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3api list-object-versions `
>> --bucket cloudsec-lab-bucket-tanmay-01 `
>> --profile admin
{
  "Versions": [
    {
      "ETag": "\"2da08c41858a1281472469b4869c1a1c\"",
      "ChecksumAlgorithm": [
        "CRC64NVME"
      ],
      "ChecksumType": "FULL_OBJECT",
      "Size": 31,
      "StorageClass": "STANDARD",
      "Key": "test.txt",
      "VersionId": "P4_6gDlRaeVhy7gelFem2SmTapBHzQ0v",
      "IsLatest": true,
      "LastModified": "2026-08-24T14:24:13+00:00",
      "Owner": {
        "ID": "eafdf07491481660b88c50baf7bb7b3622ff219ffa180a39c27d17d8d1c6337e"
      }
    }
  ],
  "RequestCharged": null,
  "Prefix": ""
}
PS C:\Users\patil\PycharmProjects\Aws-security-project>
```

Figure 6. S3 object metadata confirming AES256 encryption and versioning.

6. IAM Least Privilege

Two groups were created: Developers and Auditors. Developers were intended for read/write work on the S3 bucket, while Auditors were read-only. An unauthorized test user was used to confirm that access outside the intended permissions was denied.

```
Windows PowerShell
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws iam create-group --group-name Developers --profile admin
{
  "Group": {
    "Path": "/",
    "GroupName": "Developers",
    "GroupId": "AGPAZYXPLRF4FR4E7RKOD",
    "Arn": "arn:aws:iam::671590615416:group/Developers",
    "CreateDate": "2026-08-25T06:04:27+00:00"
  }
}
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws iam create-group --group-name Auditors --profile admin
{
  "Group": {
    "Path": "/",
    "GroupName": "Auditors",
    "GroupId": "AGPAZYXPLRF4NSU6AEJLM",
    "Arn": "arn:aws:iam::671590615416:group/Auditors",
    "CreateDate": "2026-08-25T06:05:06+00:00"
  }
}
PS C:\Users\patil\PycharmProjects\Aws-security-project>
```

Figure 7. Developers and Auditors IAM groups created for the lab.

6.1 Validation results

Identity / Test	Observed result
Auditor – list bucket	Allowed
Auditor – upload object	AccessDenied
Unauthorized – list bucket	AccessDenied
Public ACL attempt	AccessDenied

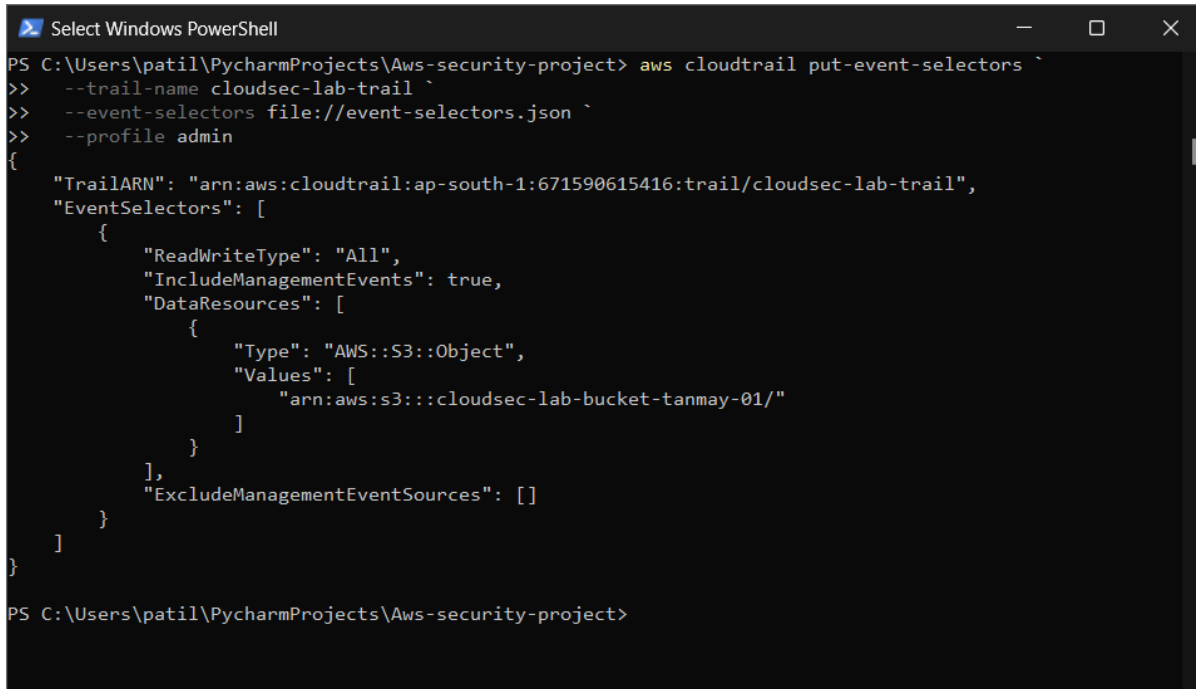
```
Windows PowerShell
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 ls s3://cloudsec-lab-bucket-tanmay-01 --profile auditor
2026-08-24 19:54:13    31 test.txt
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 ls s3://cloudsec-lab-bucket-tanmay-01/x.txt --profile auditor
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 ls s3://cloudsec-lab-bucket-tanmay-01 --profile auditor
2026-08-24 19:54:13    31 test.txt
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 cp test.txt s3://cloudsec-lab-bucket-tanmay-01/x.txt --profile auditor
upload failed: \test.txt to s3://cloudsec-lab-bucket-tanmay-01/x.txt An error occurred (AccessDenied) when calling the PutObject operation: User: arn:aws:iam::671590615416:user/test-auditor is not authorized to
perform: s3:PutObject on resource: "arn:aws:s3:::cloudsec-lab-bucket-tanmay-01/x.txt" because no identity-based policy allows the s3:PutObject action
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws s3 ls s3://cloudsec-lab-bucket-tanmay-01 --profile unauthorized
aws: [ERROR]: An error occurred (AccessDenied) when calling the ListObjectsV2 operation: User: arn:aws:iam::671590615416:user/test-unauthorized is not authorized to perform: s3:ListBucket on resource: "arn:aws:s
3:::cloudsec-lab-bucket-tanmay-01" because no identity-based policy allows the s3:ListBucket action
PS C:\Users\patil\PycharmProjects\Aws-security-project>
```

Figure 8. IAM least-privilege validation showing successful read access and denied write/unauthorized access.

7. CloudTrail and CloudWatch Monitoring

7.1 S3 data events

S3 object-level data events were enabled for the project bucket. Read/write events were captured together with management events during the testing phase.

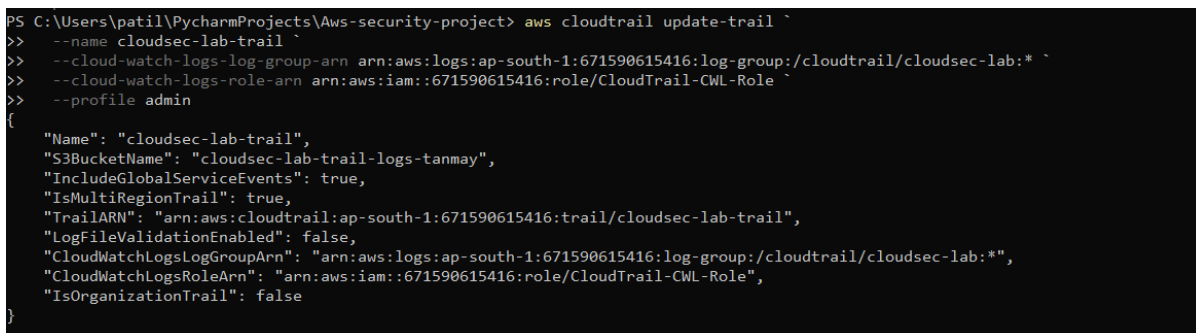


```
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws cloudtrail put-event-selectors `
>> --trail-name cloudsec-lab-trail `
>> --event-selectors file://event-selectors.json `
>> --profile admin
{
  "TrailARN": "arn:aws:cloudtrail:ap-south-1:671590615416:trail/cloudsec-lab-trail",
  "EventSelectors": [
    {
      "ReadWriteType": "All",
      "IncludeManagementEvents": true,
      "DataResources": [
        {
          "Type": "AWS::S3::Object",
          "Values": [
            "arn:aws:s3:::cloudsec-lab-bucket-tanmay-01/"
          ]
        }
      ],
      "ExcludeManagementEventSources": []
    }
  ]
}
```

Figure 9. CloudTrail event selector configuration for S3 object-level data events.

7.2 CloudTrail to CloudWatch Logs

CloudTrail was configured to deliver events to the CloudWatch Logs group /cloudtrail/cloudsec-lab using the dedicated CloudTrail-to-CloudWatch IAM role.



```
PS C:\Users\patil\PycharmProjects\Aws-security-project> aws cloudtrail update-trail `
>> --name cloudsec-lab-trail `
>> --cloud-watch-logs-log-group-arn arn:aws:logs:ap-south-1:671590615416:log-group:/cloudtrail/cloudsec-lab:* `
>> --cloud-watch-logs-role-arn arn:aws:iam::671590615416:role/CloudTrail-CWL-Role `
>> --profile admin
{
  "Name": "cloudsec-lab-trail",
  "S3BucketName": "cloudsec-lab-trail-logs-tanmay",
  "IncludeGlobalServiceEvents": true,
  "IsMultiRegionTrail": true,
  "TrailARN": "arn:aws:cloudtrail:ap-south-1:671590615416:trail/cloudsec-lab-trail",
  "LogFileValidationEnabled": false,
  "CloudWatchLogsLogGroupArn": "arn:aws:logs:ap-south-1:671590615416:log-group:/cloudtrail/cloudsec-lab:*",
  "CloudWatchLogsRoleArn": "arn:aws:iam::671590615416:role/CloudTrail-CWL-Role",
  "IsOrganizationTrail": false
}
```

Figure 10. CloudTrail configuration showing CloudWatch Logs destination and IAM role.

8. Detection Rules and Alarms

Three simple detection rules were implemented. Each matching event increments a custom CloudWatch metric, and each metric is connected to an alarm with SNS notification.

Detection	Metric	Alarm
DeleteBucket	DeleteBucketCount	DeleteBucketAlarm
PutBucketAcl	PutBucketAclCount	PutBucketAclAlarm
ConsoleLogin without MFA	ConsoleLoginNoMFACount	ConsoleLoginNoMFAAlarm

```
Windows PowerShell
PS C:\Users\patil> aws logs describe-metric-filters --log-group-name /cloudtrail/cloudsec-lab --profile admin
{
  "metricFilters": [
    {
      "filterName": "ConsoleLoginNoMFAFilter",
      "filterPattern": "{ $.$eventName = ConsoleLogin } && { $.$additionalEventData.MFAUsed != Yes }",
      "metricTransformations": [
        {
          "metricName": "ConsoleLoginNoMFACount",
          "metricNamespace": "CloudSecLab",
          "metricValue": "1"
        }
      ]
    },
    {
      "filterName": "DeleteBucketFilter",
      "filterPattern": "{ $.$eventName = DeleteBucket }",
      "metricTransformations": [
        {
          "metricName": "DeleteBucketCount",
          "metricNamespace": "CloudSecLab",
          "metricValue": "1"
        }
      ]
    },
    {
      "filterName": "PutBucketAclFilter",
      "filterPattern": "{ $.$eventName = PutBucketAcl }",
      "metricTransformations": [
        {
          "metricName": "PutBucketAclCount",
          "metricNamespace": "CloudSecLab",
          "metricValue": "1"
        }
      ]
    }
  ]
}
```

Figure 11. CloudWatch metric filters showing all three detection rules.

8.1 Alert evaluation

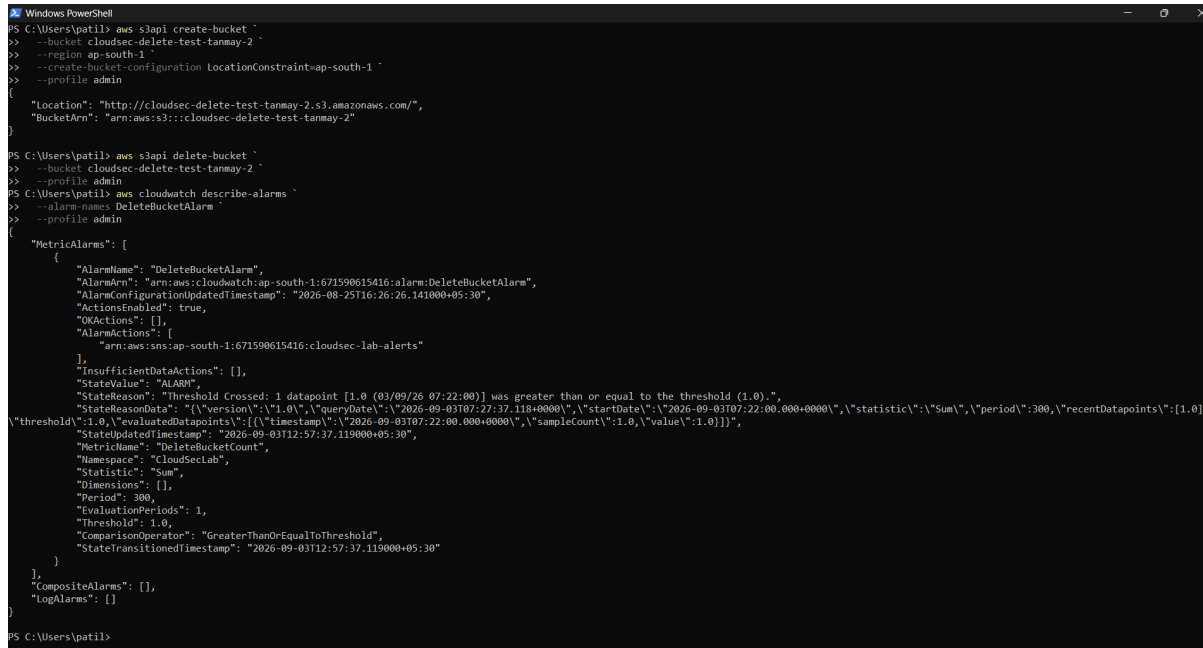
The alarms were configured with a threshold of 1 matching event in a 300-second period. The resulting alarm states were validated after the corresponding activities were performed.

9. Security Validation

The lab was tested with deliberate actions intended to represent common security-relevant activity. The tests were performed against the lab resources and were followed by CloudWatch/SNS validation.

9.1 DeleteBucket detection

A separate temporary bucket named `cloudsec-delete-test-tanmay-2` was created specifically for the DeleteBucket test. The temporary bucket—not the main project bucket—was deleted, generating the DeleteBucket event. The main bucket was preserved until the final cleanup stage.



```
PS C:\Users\patil> aws s3api create-bucket `
> --bucket cloudsec-delete-test-tanmay-2 `
> --region ap-south-1 `
> --create-bucket-configuration LocationConstraint=ap-south-1 `
> --profile admin
{
  "Location": "http://cloudsec-delete-test-tanmay-2.s3.amazonaws.com/",
  "BucketArn": "arn:aws:s3:::cloudsec-delete-test-tanmay-2"
}

PS C:\Users\patil> aws s3api delete-bucket `
> --bucket cloudsec-delete-test-tanmay-2 `
> --profile admin
PS C:\Users\patil> aws cloudwatch describe-alarms `
> --alarm-names DeleteBucketAlarm `
> --profile admin
{
  "MetricAlarms": [
    {
      "AlarmName": "DeleteBucketAlarm",
      "AlarmArn": "arn:aws:cloudwatch:ap-south-1:671590615416:alarm:DeleteBucketAlarm",
      "AlarmConfigurationUpdatedTimestamp": "2026-08-25T16:26:26.141000+05:30",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:sns:ap-south-1:671590615416:cloudsec-lab-alerts"
      ],
      "InsufficientDataActions": [],
      "StateValue": "ALARM",
      "StateReason": "Threshold Crossed: 1 datapoint [1.0 (03/09/26 07:22:00)] was greater than or equal to the threshold (1.0).",
      "StateReasonData": "{\n\"version\": \"1.0\", \"queryDate\": \"2026-09-03T07:27:37.118+0000\", \"startDate\": \"2026-09-03T07:22:00.000+0000\", \"statistic\": \"Sum\", \"period\": 300, \"recentDatapoints\": [{1.0}, {\n\"threshold\": 1.0, \"evaluatedDatapoints\": [{\n\"timestamp\": \"2026-09-03T07:22:00.000+0000\", \"sampleCount\": 1.0, \"value\": 1.0}]\n\", \"StateUpdatedTimestamp\": \"2026-09-03T12:57:37.119000+05:30\", \"MetricName\": \"DeleteBucketCount\", \"Namespace\": \"cloudseclab\", \"Statistic\": \"Sum\", \"Dimensions\": [], \"Period\": 300, \"EvaluationPeriods\": 1, \"Threshold\": 1.0, \"ComparisonOperator\": \"GreaterThanOrEqualToThreshold\", \"StateTransitionedTimestamp\": \"2026-09-03T12:57:37.119000+05:30\" }\n]",
      "StateUpdatedTimestamp": "2026-09-03T12:57:37.119000+05:30",
      "MetricName": "DeleteBucketCount",
      "Namespace": "cloudseclab",
      "Statistic": "Sum",
      "Dimensions": [],
      "Period": 300,
      "EvaluationPeriods": 1,
      "Threshold": 1.0,
      "ComparisonOperator": "GreaterThanOrEqualToThreshold",
      "StateTransitionedTimestamp": "2026-09-03T12:57:37.119000+05:30"
    }
  ],
  "CompositeAlarms": [],
  "LogAlarms": []
}
```

Figure 12. DeleteBucketAlarm reaching ALARM after the temporary test bucket was deleted.

9.2 PutBucketAcl detection

A public-read ACL change was attempted against the main bucket. S3 Block Public Access prevented the change, while the activity was still useful as a security-relevant event for monitoring and detection.

```
Windows PowerShell
PS C:\Users\patil> aws s3api put-bucket-acl
PS C:\Users\patil> --bucket cloudsec-lab-bucket-tammy-01
PS C:\Users\patil> --acl public-read
PS C:\Users\patil> --profile admin

aws: [ERROR] An error occurred (AccessDenied) when calling the PutBucketAcl operation: User: arn:aws:iam::671590615416:user/tammy-admin is not authorized to perform: s3:PutBucketAcl on resource: "arn:aws:s3:::cloudsec-lab-bucket-tammy-01" because public ACLs are prevented by the BlockPublicAcls setting in S3 Block Public Access.
PS C:\Users\patil> aws cloudwatch describe-alarms
PS C:\Users\patil> --alarm-names PutBucketAclAlarm
PS C:\Users\patil> --profile admin

{
  "MetricAlarms": [
    {
      "AlarmName": "PutBucketAclAlarm",
      "AlarmArn": "arn:aws:cloudwatch:ap-south-1:671590615416:alarm:PutBucketAclAlarm",
      "AlarmConfigurationUpdatedTimestamp": "2026-08-25T16:26:53.653000+05:30",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:sns:ap-south-1:671590615416:cloudsec-lab-alerts"
      ],
      "InsufficientDataActions": [],
      "StateValue": "ALARM",
      "StateReason": "Threshold Crossed: 1 datapoint [1.0 (03/09/26 07:14:00)] was greater than or equal to the threshold (1.0).",
      "StateReasonData": "{\n\"version\": \"1.0\", \"queryDate\": \"2026-09-03T07:19:05.574+0000\", \"startDate\": \"2026-09-03T07:14:00.000+0000\", \"statistic\": \"Sum\", \"period\": 300, \"recentDatapoints\": [1.0], \"threshold\": 1.0, \"evaluatedDatapoints\": [{\"timestamp\": \"2026-09-03T07:14:00.000+0000\", \"sampleCount\": 1.0, \"value\": 1.0}]",
      "StateUpdatedTimestamp": "2026-09-03T12:49:05.576000+05:30",
      "MetricName": "PutBucketAclCount",
      "Namespace": "CloudSecLab",
      "Statistic": "Sum",
      "Dimensions": [],
      "Period": 300,
      "EvaluationPeriods": 1,
      "Threshold": 1.0,
      "ComparisonOperator": "GreaterThanOrEqualToThreshold",
      "StateTransitionedTimestamp": "2026-09-03T12:49:05.576000+05:30"
    }
  ],
  "CompositeAlarms": [],
  "LogAlarms": []
}
```

Figure 13. PutBucketAclAlarm reaching ALARM after the public ACL attempt.

9.3 Console login without MFA

A console login without MFA was used to validate the no-MFA detection rule. The associated alarm moved into ALARM.

```
Windows PowerShell
PS C:\Users\patil> aws cloudwatch describe-alarms --alarm-names ConsoleLoginNoMFAAlarm --profile admin

{
  "MetricAlarms": [
    {
      "AlarmName": "ConsoleLoginNoMFAAlarm",
      "AlarmArn": "arn:aws:cloudwatch:ap-south-1:671590615416:alarm:ConsoleLoginNoMFAAlarm",
      "AlarmConfigurationUpdatedTimestamp": "2026-08-25T16:27:22.643000+05:30",
      "ActionsEnabled": true,
      "OKActions": [],
      "AlarmActions": [
        "arn:aws:sns:ap-south-1:671590615416:cloudsec-lab-alerts"
      ],
      "InsufficientDataActions": [],
      "StateValue": "INSUFFICIENT_DATA",
      "StateReason": "Insufficient Data: 1 datapoint was unknown.",
      "StateReasonData": "{\n\"version\": \"1.0\", \"queryDate\": \"2026-09-03T08:04:28.231+0000\", \"startDate\": \"2026-09-03T08:04:28.231+0000\", \"statistic\": \"Sum\", \"period\": 300, \"recentDatapoints\": [], \"threshold\": 1.0, \"evaluatedDatapoints\": [{\"timestamp\": \"2026-09-03T07:59:00.000+0000\"}]",
      "StateUpdatedTimestamp": "2026-09-03T13:34:28.233000+05:30",
      "MetricName": "ConsoleLoginNoMFAAccount",
      "Namespace": "CloudSecLab",
      "Statistic": "Sum",
      "Dimensions": [],
      "Period": 300,
      "EvaluationPeriods": 1,
      "Threshold": 1.0,
      "ComparisonOperator": "GreaterThanOrEqualToThreshold",
      "StateTransitionedTimestamp": "2026-09-03T13:34:28.233000+05:30"
    }
  ],
  "CompositeAlarms": [],
  "LogAlarms": []
}
```

Figure 14. ConsoleLoginNoMFAAlarm reaching ALARM.

10. SNS Alerting

Each CloudWatch alarm was connected to the cloudsec-lab-alerts SNS topic. After the alarm state changed to ALARM, AWS Notifications emails were received for each detection.

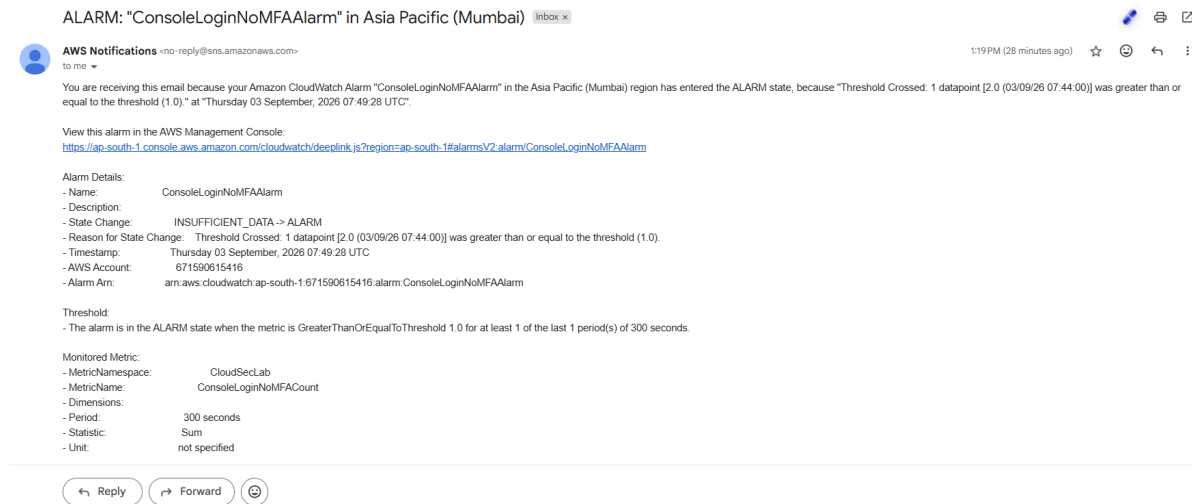


Figure 15. SNS email notification for the ConsoleLoginNoMFA alarm.

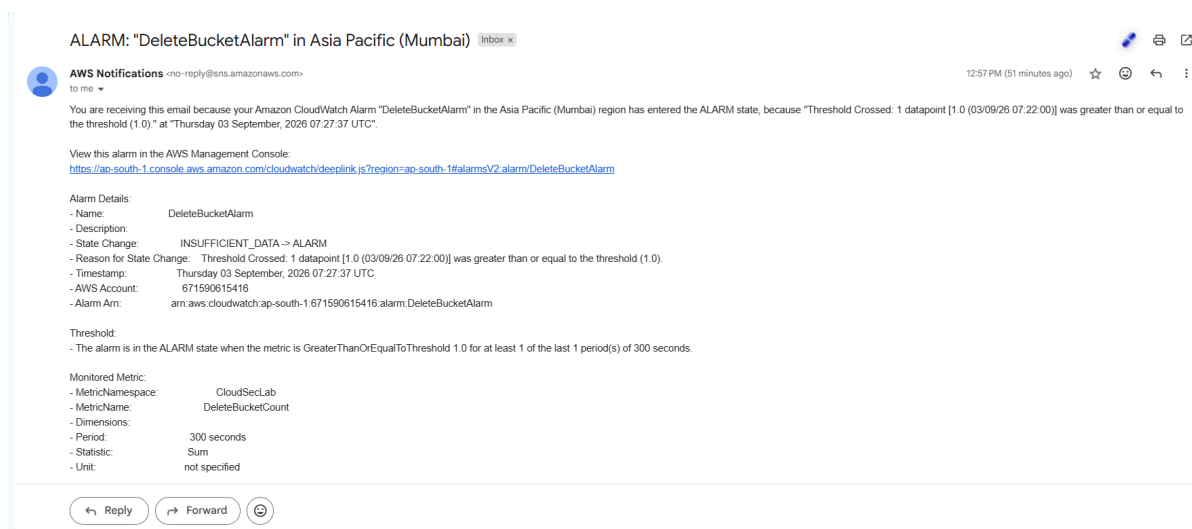


Figure 16. SNS email notification for the DeleteBucket alarm.

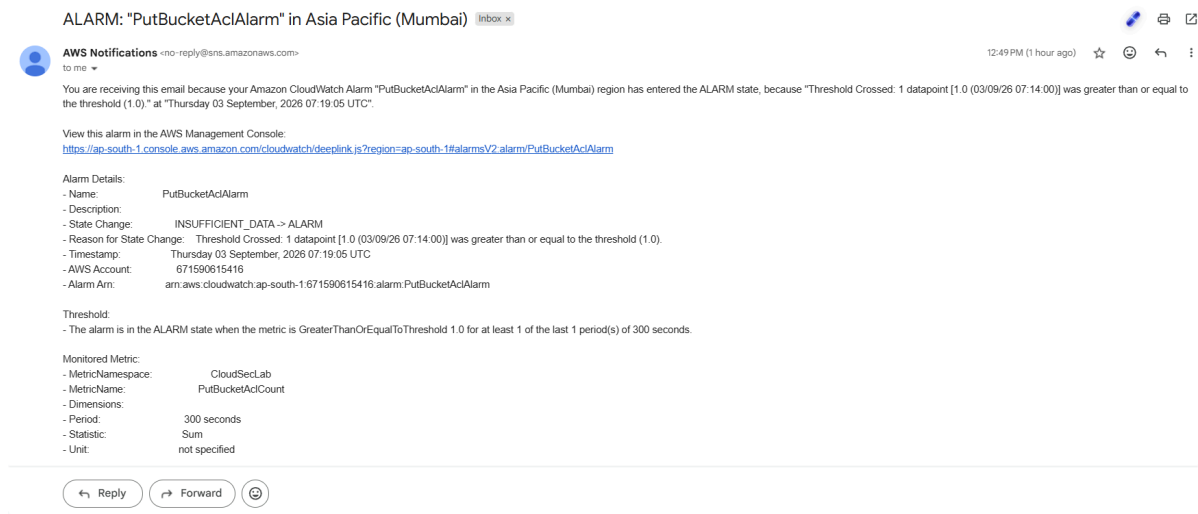


Figure 17. SNS email notification for the PutBucketAcl alarm.

The three notifications provide the final alerting evidence: ConsoleLoginNoMFAAlarm, DeleteBucketAlarm, and PutBucketAclAlarm were all delivered by SNS after their corresponding CloudWatch alarms entered ALARM.

11. Python Log Analyzer

A local Python analyzer was added as an extra analysis layer. It uses boto3 to query recent CloudWatch Logs and pandas to structure the results. The analyzer identifies selected sensitive actions and flags unusually high event volume from a single source IP.

11.1 Detection logic

- HIGH: DeleteBucket
- HIGH: PutBucketAcl
- HIGH: DeleteObject
- MEDIUM: more than 20 events from the same source IP within the analysis window

11.2 Actual analyzer result

The completed run produced multiple HIGH findings and a MEDIUM high-volume finding. The observed output included DeleteObject, PutBucketAcl, and DeleteBucket activity, plus 27 events from one source IP within the analysis window.

```
Windows PowerShell
delete: s3://cloudsec-lab-bucket-tanmay-01/analyzer-test.txt
PS C:\Users\patil\PycharmProjects\Aws-security-project> & C:\Users\patil\AppData\Local\Programs\Python\Python312\python.exe ^
>> C:\Users\patil\PycharmProjects\Aws-security-project\analyzer.py
[HIGH] Sensitive action DeleteObject by arn:aws:iam::671590615416:user/tanmay-admin from 206.84.235.141
[HIGH] Sensitive action PutBucketAcl by arn:aws:iam::671590615416:user/tanmay-admin from 206.84.235.141
[HIGH] Sensitive action DeleteBucket by arn:aws:iam::671590615416:user/tanmay-admin from 206.84.235.141
[MEDIUM] High call volume from 206.84.235.141: 27 events in window
PS C:\Users\patil\PycharmProjects\Aws-security-project>
```

Figure 18. Actual output from analyzer.py showing HIGH sensitive-action findings and a MEDIUM volume finding.

11.3 Dependency and credential handling

The analyzer runs under Python 3.12 with boto3 and pandas installed in the same interpreter. boto3 was configured to use the AWS CLI admin profile for the lab. The final script uses a timezone-aware UTC timestamp calculation.

12. Final Cleanup and Decommissioning

After evidence collection and testing, the AWS lab resources were decommissioned. The cleanup was performed in dependency order to avoid leaving active monitoring or test identities behind.

Resource	Final status
CloudWatch alarms	Deleted
CloudWatch metric filters	Deleted
CloudWatch log group	Deleted
SNS topic	Deleted
CloudTrail trail	Deleted
Main S3 lab bucket	Deleted after removing versions/delete markers
Test IAM users	Deleted
Developers / Auditors groups	Deleted
tanmay-admin user	Retained
Lab admin access key	Deleted after AWS work was finished

The final verification showed no remaining lab alarms, log group, SNS topic, CloudTrail trail, test users, Developers/Auditors groups, or cloudsec lab buckets. The tanmay-admin IAM user was intentionally retained.

13. Repository Structure

The GitHub repository is intentionally small. Each folder has a single purpose and there is no unnecessary infrastructure-as-code or application scaffolding.

```
aws-secure-cloud-storage/  
├── README.md  
├── policies/  
│   ├── bucket-policy.json  
│   ├── developers-policy.json  
│   ├── auditors-policy.json  
│   ├── cloudtrail-cwl-trust-policy.json  
│   └── cloudtrail-cwl-permissions.json  
├── scripts/  
│   ├── analyzer.py  
│   └── cleanup.sh  
├── tests/  
│   └── validation-matrix.md  
├── evidence/  
│   ├── architecture.png  
│   ├── environment-admin-identity.png  
│   ├── environment-cloudtrail-active.png  
│   ├── s3-public-access-block.png  
│   ├── s3-https-only-validation.png  
│   ├── s3-encryption-versioning.png  
│   ├── iam-groups.png  
│   ├── iam-least-privilege-validation.png  
│   ├── cloudtrail-s3-data-events.png  
│   ├── cloudtrail-cloudwatch-integration.png  
│   ├── detection-metric-filters.png  
│   ├── alarm-triggered-delete-bucket.png  
│   ├── alarm-triggered-put-bucket.png  
│   ├── alarm-triggered-no-mfa.png  
│   ├── sns-security-alert-delete-bucket.png  
│   ├── sns-security-alert-put-bucket.png  
│   ├── sns-security-alert-no-mfa.png  
│   └── analyzer-output.png  
└── docs/  
    └── final-report.pdf
```

13.1 Purpose of each folder

Folder / File	Purpose
README.md	Concise project overview for GitHub visitors.
policies/	Actual IAM, S3, and CloudTrail policy JSON used in the lab.
scripts/	Python log analyzer and cleanup script.
tests/	Small validation matrix summarizing security tests.
evidence/	Selected screenshots and actual analyzer output.
docs/	Detailed final project documentation/report.

14. Limitations and Notes

- This is a controlled security lab, not a production AWS deployment.
- The lab used a long-lived IAM access key for convenience; it was deleted during final cleanup.
- The detection layer uses CloudWatch Metric Filters and Alarms rather than a full SIEM/SOAR platform.
- S3 data events were enabled during testing and disabled during cleanup to avoid unnecessary ongoing data-event collection.
- The Python analyzer is intentionally small and is an additional visibility/validation tool, not a complete anomaly-detection engine.
- Production environments would normally use stronger centralized identity, short-lived credentials, and broader managed security services.

14.1 Evidence publication hygiene

Before making the screenshots public on GitHub, account identifiers, full AWS ARNs, and any other unnecessary account-specific information should be redacted. No access keys, passwords, session tokens, or other credentials should be committed to the repository.

14.2 Implementation Observations and Lessons Learned

The following observations were recorded from the actual implementation and troubleshooting of this lab. They describe what was learned while building, testing, and cleaning up the environment.

1. Service permissions and resource policies

AWS services do not automatically gain access to a resource simply because the resources are in the same account. When a service principal needs access, the required permissions must be provided. In this lab, CloudTrail log delivery required the appropriate destination-bucket permissions, while CloudTrail-to-CloudWatch delivery required the IAM role and permissions configured for that integration. Because the resources were created separately, those permissions had to be configured explicitly.

2. Resource creation order can reduce manual setup

A simpler lab build can plan the CloudTrail logging destination and its required permissions before the rest of the storage environment is finalized. In this project, the resources were created separately, so the required policy and role configuration had to be handled manually.

3. AWS CLI names are case-sensitive

AWS CLI resource names and values must match exactly. A capitalization mistake in names such as Developers or Auditors was enough to cause commands to fail. Copying exact resource names before troubleshooting avoids this class of error.

4. PowerShell is different from Bash

The project was performed mainly in Windows PowerShell. Bash-style habits, quoting, line continuation, and path syntax do not always translate directly. The cleanup stage also exposed this difference when a Unix-style path was interpreted incorrectly by Git Bash on Windows.

5. CloudTrail management events and S3 data events are different

Management events cover administrative activity such as creating resources or changing configuration. S3 object-level activity requires data-event configuration. In this lab, S3 data events were enabled temporarily for testing and disabled during cleanup.

6. Versioned S3 buckets require special cleanup

With versioning enabled, a normal object delete can leave delete markers or older versions behind. The main bucket therefore had to be checked with list-object-versions and cleared before the bucket itself could be deleted.

7. AWS monitoring resources are regional

During testing, an alarm that had triggered was not visible when it was queried from the wrong region. Checking the same resource in ap-south-1 resolved the confusion. The practical lesson is to keep the AWS region consistent when creating and reviewing regional services such as CloudWatch alarms.

8. SNS does not email for every repeated match

The SNS action was attached to the CloudWatch alarm state change. Once an alarm remained in ALARM, additional matching events did not necessarily produce another email for every event. This explained why CloudWatch activity could continue to increase while the number of SNS emails remained lower.

Summary of practical lessons

Area	Observed lesson
Permissions	Service access requires explicit permissions; separate resources may require manual policy/role setup.
CLI usage	AWS CLI resource names and values are case-sensitive.
Shell	PowerShell syntax differs from Bash for quoting, paths, and line continuation.
Logging	Management events and S3 data events are separate CloudTrail controls.
S3 versioning	Delete markers and versions must be removed before a versioned bucket can be deleted.
Regions	CloudWatch alarms are regional and must be checked in the correct AWS region.
SNS	Notifications are tied to alarm state changes; repeated matching events while an alarm stays ALARM do not necessarily produce a new email each time.

15. Final Outcome

The completed lab demonstrated a complete, practical security workflow around an AWS S3 data store:

Secure → Monitor → Detect → Alert → Analyze

S3 was protected with access, encryption, versioning, and transport controls. IAM enforced least privilege. CloudTrail recorded activity and CloudWatch processed it. Three detection rules triggered alarms and SNS emails. The Python analyzer then identified sensitive and high-volume activity from the collected events.

The project was subsequently decommissioned, leaving the AWS account and administrative identity intact while removing the lab-specific resources and credentials used for testing.

Appendix A — Selected Validation Matrix

Test	Expected	Result
Auditor: List bucket	Allowed	PASS
Auditor: Upload object	Denied	PASS
Unauthorized: List bucket	Denied	PASS
Public ACL attempt	Denied	PASS
HTTP access	Denied (403)	PASS
DeleteBucket detection	Alarm	PASS
PutBucketAcl detection	Alarm	PASS
No-MFA login detection	Alarm	PASS
Python analyzer	Sensitive activity detected	PASS